

Effect of Women's Nutrition before and during Early Pregnancy on Maternal and Infant Outcomes: A Systematic Review

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Abstract

Current understanding of biologic processes indicates that women's nutritional status before and during early pregnancy may play an important role in determining early developmental processes and ensuring successful pregnancy outcomes. We conducted a systematic review of the evidence for the impact of maternal nutrition before and during early pregnancy (<12 weeks gestation) on maternal, neonatal and child health outcomes and included 45 articles (nine intervention trials and 32 observational studies) that were identified through PubMed and EMBASE database searches and examining review articles. Intervention trials and observational studies show that periconceptional (<12 weeks gestation) folic acid supplementation significantly reduced the risk of neural tube defects. Observational studies suggest that preconceptional and periconceptional intake of vitamin and mineral supplements is associated with a reduced risk of delivering offspring who are low birthweight and/or small-for-gestational age (SGA) and preterm deliveries (PTD). Some studies report that indicators of maternal prepregnancy size, low stature, underweight and overweight are associated with increased risks of PTD and SGA. The available data indicate the importance of women's nutrition prior to and during the first trimester of pregnancy, but there is a need for well-designed prospective studies and controlled trials in developing country settings that examine relationships with low birthweight, SGA, PTD, stillbirth and maternal and neonatal mortality. The knowledge gaps that need to be addressed include the evaluation of periconceptional interventions such as food supplements, multivitamin-mineral supplements and/or specific micronutrients (iron, zinc, iodine, vitamin B-6 and B-12) as well as the relationship between measures of prepregnancy body size and composition and maternal, neonatal and child health outcomes.

Keywords: *early pregnancy, women's nutrition, birth outcomes.*

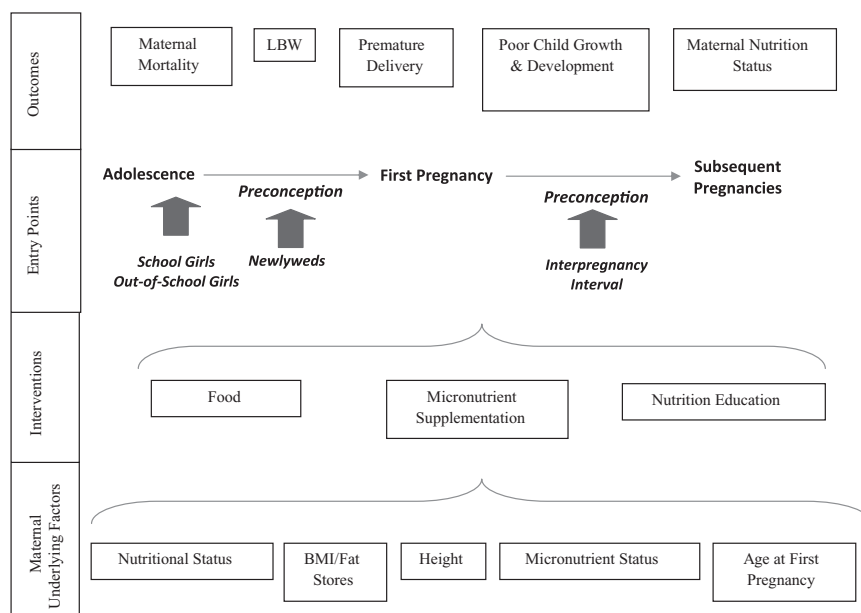
Women's nutrition, before and during pregnancy, may play a key role in reproductive health and is recognised as being important for optimising pregnancy outcomes.^{1,2} The availability and supply of nutrients to the developing fetus depends on maternal nutritional status which in turn depends on her nutrient stores, dietary intake and obligatory requirements. Most of the studies that have examined the importance of nutrition during pregnancy typically focus on the second and/or the third trimester by which time key processes such as organogenesis have been completed.³ Women's nutritional status just before conception and/or during early pregnancy (<12 weeks gestation), when women are typically unaware of their

pregnancy status, may influence pregnancy outcomes by affecting critical developmental processes that begin early in pregnancy as well as the availability of nutrients. Animal studies suggest that periconceptional undernutrition may influence the hypothalamic-pituitary-adrenal axis which in turn influences outcomes such as pre-eclampsia and preterm delivery (PTD).⁴ Ensuring an adequate supply of nutrients to the fetus throughout gestation also depends on placental function which is determined in early pregnancy and may be influenced by maternal nutrition during early pregnancy.^{3,5} Maternal endocrine and metabolic responses that occur early in pregnancy in turn influence the supply and utilisation of available nutrients for the rapidly growing fetus later in pregnancy.^{6,7}

Various aspects of maternal nutrition that are particularly relevant for the developing world and may influence pregnancy outcomes are shown in Figure 1.

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Figure 1. Conceptual framework of stages of pregnancy potentially impacted by nutrition. BMI, body mass index; LBW, low birthweight.



Women living in resource poor settings are often malnourished before pregnancy; they may be short as a result of early childhood malnutrition, and underweight and anaemic due to inadequate food intakes and infections. In some settings, overweight and obesity are also emerging concerns due to poor diet.^{8,9} Several observational studies have shown that measures of body size such as height, weight and body mass index (BMI) are associated with adverse birth outcomes such as low birthweight (LBW) and small-for-gestational age (SGA), although the exact timing of body measurement is unclear.^{10–12} Age at the time of conception and duration of inter-pregnancy interval are also important as they may influence the availability of nutrients at the time of conception and during early pregnancy. Adolescent girls who have not completed their own growth and development may be at increased risk of being shorter, lighter and/or depleted stores of energy and micronutrients such as iron, iodine and vitamin A; women with short inter-pregnancy intervals may also be at increased risk of nutrient deficiencies in resource poor settings.^{13,14} Various nutrients may influence pregnancy outcomes by altering both maternal and fetal metabolism due to their roles in modulating oxidative stress, enzyme function, signal transduction and transcription pathways that occur early in pregnancy,^{3,15} namely during the critical periods of preconception, conception, implantation, placentation and embryo- or organogen-

esis. Nutrients such as iron, zinc, iodine and long chain n-3 polyunsaturated fatty acids (LCPUFA) play critical roles in development of the brain and nervous system, whereas vitamins A, B-6, B-12 and folic acid influence oxidative pathways and methylation.

Nutrition during early pregnancy may affect placental function, which has been associated with adverse pregnancy outcomes such as pre-eclampsia, PTD and fetal growth restriction. Proposed mechanisms include lowered number and surface area of arterioles in tertiary villi and reduction in spiral artery formation as a result of impaired function of trophoblasts due to oxidative stress and/or inflammation.⁵ LCPUFA and iron status during early pregnancy have been shown to be inversely associated with placental weight and surface area of capillaries involved in gas exchange, respectively.^{16,17} Several micronutrients can also influence inflammation and oxidative stress early in pregnancy; vitamins A and D, zinc and fatty acids may influence immune function whereas vitamins C, E, B-6, B-12 and folic acid may reduce oxidative damage to the placenta. Nutrients such as vitamins A, B-6, B-12 and folic acid and zinc also affect embryogenesis that occurs early in pregnancy and may be related to pregnancy loss and fetal malformations. These nutrients are involved in several biochemical pathways such as the homocysteine pathway and influence processes such as methylation which in turn affects cell replication and differentiation.⁵ The most

well-studied effect of periconceptual nutrition is the protective effect of folic acid in the first 28 days of pregnancy in reducing the risk of delivering infants with neural tube defect (NTD), which contribute to significant mortality and morbidity.^{18–20} Much less is known about other nutrients.

The objective of this paper, which is part of a series on the role of maternal nutrition for improving maternal, neonatal and child health outcomes (MNCH) that are significant public health problems in many developing countries, is to conduct a systematic review of the evidence on the role of nutrition before and during early pregnancy on maternal morbidity and mortality, pregnancy loss including stillbirths, birth defects, LBW, PTD and infant mortality.

Methods

Search strategy

We identified published studies using PubMed and EMBASE search engines. The search was carried out by combining the results of three separate strategies using the following key concepts and related key words: (1) preconception and periconception terms ('preconception' or 'periconception' or 'prepregnancy' or 'pre-pregnancy' or 'before pregnancy'), (2) nutritional terms ('intervention' or 'nutrition' or 'micro-nutrient supplementation' or 'food fortification' or 'body mass index' or 'weight' or 'multivitamin' or 'multivitamin-mineral' or 'vitamin' or 'iodine' or 'zinc' or 'folic acid' or 'iron'), and (3) maternal, neonatal and child health outcomes of interest ('low birth weight' or 'birth size' or 'birth weight' or 'intrauterine growth restriction' or 'preterm birth' or 'preterm delivery' or 'gestational age' or 'morbidity' or 'mortality' or 'growth' or 'nutritional status' or 'stillbirth' or 'pregnancy loss' or 'birth defects' and 'neonatal' or 'infant' or 'child' or 'children' or 'maternal'). We included all studies from 1950 to July 2011 with no language restriction but limited to 'humans'. Additional studies were identified through hand search of references from previous review articles. The inclusion and exclusion criteria were as follows:

Inclusion criteria: (1) intervention and observational studies, (2) nutrition exposure (intervention or indicators of nutritional status) was measured right before (within 1 year of conception) and/or during early pregnancy (<12 weeks gestation), (3) studies that included MNCH outcomes (namely, maternal morbidity

and mortality, pregnancy loss, stillbirths, birth defects, birth size, PTD, neonatal and infant mortality), and (4) appropriate comparison group.

Exclusion criteria: (1) animal studies, (2) nutrition exposure was assessed or began at later stages of pregnancy (beyond first trimester), (3) non-nutritional studies, (4) review articles, and (5) poorly defined comparison group (for, e.g. in programme evaluations, intervention of interest continued during pregnancy).

Data abstraction

The abstracts of all potential publications were reviewed independently by two co-authors (T. G. and A. Z.) initially to identify eligible publications for data abstraction. The senior author (U. R.) reviewed publications that were identified for inclusion by only one co-author to determine eligibility. Relevant study attributes (qualitative and quantitative) were abstracted from the selected publications using standardised forms developed for the overall project by one co-author (T. G.) and reviewed for accuracy by another co-author (F. G.) and the senior author (U. R.). We assessed the overall quality of evidence for the outcomes by the Grading of Recommendations, Assessment, Development and Evaluation criteria.^{21,22}

Results

A total of 441 titles were identified on PubMed and EMBASE searches, of which we carefully reviewed 62 articles that included five that appeared in the review articles but not in the PubMed search (Figure 2). Careful examination of these studies resulted in the inclusion of 45 articles^{20,23–66} most of which were based on observational studies (Table 1). The main reasons for excluding articles after completing abstraction were: (i) the intervention and/or exposure occurred after 12 weeks gestation,^{67–74} (ii) missing our outcomes of interest,^{75–83} and (iii) did not measure periconceptual exposures.^{84,85}

We identified six intervention trials^{20,30,33,38,39,42,44,62,76} and seven observational studies^{26,29,34,36,43,48,51,52} that examined the relationship between preconception and/or periconception maternal nutritional status and maternal morbidity and pregnancy loss including stillbirth. In the case of child outcomes, we included nine intervention trials^{20,30–33,38,39,42,44,62} and 28 observational studies^{23–28,34,35,37,40,41,43,45–50,53–61,63–66} that examined the effects on birth defects, birth size and PTD. We

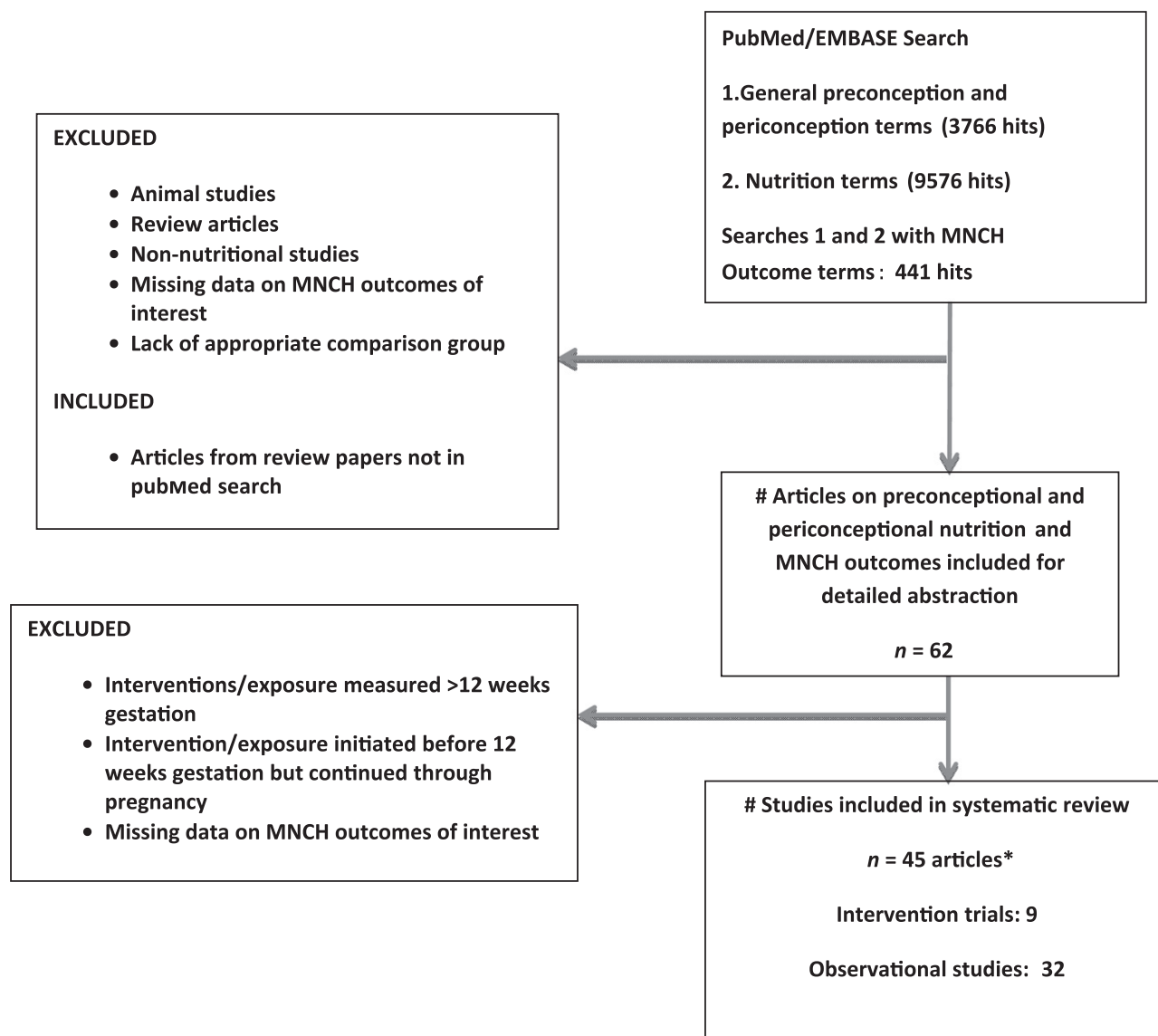


Figure 2. Studies excluded and included in the review of the preconception nutrition and pregnancy outcomes. *Some studies had more than one publication. MNCH, maternal, neonatal and child health outcomes.

did not find any studies that examined maternal or neonatal mortality. The key findings are described by outcome in the following sections and additional details are available on request.

Maternal health

Four observational studies^{26,29,43,48} examined the effects of prepregnancy and/or periconceptional nutrition on the risk of developing pre-eclampsia later in pregnancy. Phithakwatchara and Titapant⁴⁸ reported that the risk of pre-eclampsia was significantly increased [odds ratio (OR): 3.87 [95% confidence interval (CI)

2.09, 7.25]] in overweight Thai women (pre-pregnancy BMI ≥ 27 kg/m²) compared with normal weight women (BMI of 20–25 kg/m²), after adjusting for the confounding factors. This study used data from a retrospective review of medical records of the pregnant women who were at risk of gestational diabetes. Similar findings were reported in a large study of singleton nulliparous pregnancies delivered in three hospitals in Shenyang, China using data obtained from medical records.⁴³ Compared with normal weight women ($18.5 \leq \text{BMI} < 24$ kg/m²), overweight ($24 \leq \text{BMI} < 28$ kg/m²) and obese women (BMI ≥ 28 kg/m²) had significantly increased risks [adjusted

Table 1. Description of intervention and observational studies of periconceptual nutrition and maternal and child health outcomes

Reference	Design, <i>n</i>	Location	Population	Intervention/exposure	Outcomes
Intervention trials					
Berry <i>et al.</i> , 1999 ²⁰	Community-based intervention, 247 831	China	Women preparing for marriage who underwent a physical examination and were later registered with a monitoring system that records prenatal care and delivery	Daily supplement containing 400 µg folic acid	NTD ^a
Chaouki and Benmiloud, 1994 ³⁰	Community-based intervention, 1536	Algeria	Women who were planning to be pregnant	Oral iodised oil 1–3 months before conception or during the first trimester	Stillbirths, BW ^a , PTD
Chen <i>et al.</i> , 2008 ³¹	Community-based intervention, 52 043 RCT, 7540	China	Resident women planning a pregnancy who volunteered to participate in follow-up	Daily multivitamin supplement, containing 23 vitamins and microelements (including folic acid)	NTD ^b
Czeizel <i>et al.</i> , 1992, 1994 ^{32,33}		Hungary	Women (<35 years) planning a pregnancy (in most cases their first) who qualified for the Hungarian Family Planning Program trial in 1984	Vitamin supplement (containing 12 vitamins including 0.8 mg of folic acid, four minerals and three trace elements) or a trace element supplement (containing copper, manganese, zinc and a very low dose of vitamin C) daily for ≥1 month before conception and until the date of the second missed menstrual period or later	NTD ^b , BW, GA, LBW, PTD, early pregnancy loss, stillbirth
Indian Council of Medical Research Collaborating Centres and Central Technical Co-ordinating Unit, 2000 ³⁸	Double-blind, RCT, 1466	India	Non-pregnant women who had previously given birth to a child with open NTD and planned to have another child	Multivitamin supplement (ferrous sulphate, calcium phosphate, vitamins A, D, B-1, B-2, B-6 and C, nicotinamide, zinc and folic acid) or placebo (ferrous sulphate and calcium sulphate) for ≥1 month prior to conception up to 3 month after conception	Recurrence of NTD ^b , early pregnancy loss, stillbirth
Kirke <i>et al.</i> , 1992 ³⁹	Double-blind, RCT, 354	Ireland	Non-pregnant women who had previously given birth to a child with an NTD and planned to have another child	Folic acid supplement, multivitamins without folic acid, or folic acid plus multivitamins for ≥2 month prior to conception until the third missed menstrual period	Recurrence of NTD ^a
Laurence <i>et al.</i> , 1981 ⁴²	Double-blind, RCT, 111	Wales	Non-pregnant women (<35 years) who had had a previous pregnancy complicated by a fetal NTD	4 mg folic acid/day or placebo from the time contraceptive use was stopped until 6 weeks after a missed menstrual period	Recurrence of NTD ^a
Medical Research Council Vitamin Study Research Group, 1991 ⁴⁴	Multicenter randomised double-blind prevention trial, 1817	UK	Women at high risk of having a pregnancy with an NTD, because of a previous affected pregnancy	Folic acid, other vitamins, both or neither	NTD ^a
Vergel <i>et al.</i> , 1990 ⁶²	Community-based intervention, 81	Cuba	Non-pregnant women with a history of a previous NTD birth	5 mg folic acid/day for not less than one menstrual period before conception until the 10th week of pregnancy	Recurrence of NTD ^a
Observational studies					
Bitisko <i>et al.</i> , 2007 ²³	Case-control, 395	US	Cases: mothers of infants/fetuses with ≥2 unrelated major birth defects affecting ≥2 different organ systems; Controls: random sample of liveborn, non-malformed infants born in the same time period to residents of the same geographical area	Self-report of vitamin supplementation including ≥4000 µg folic acid ≥3 times/week periconceptionally (3 months before and after conception) or during the first trimester	Multiple congenital anomalies
Bower and Stanley, 1992 ²⁵	Case-control, 154	Australia	Cases: women who had infants/fetuses with malformations other than NTDs or mothers of liveborn infants without malformations	Self-report of vitamin supplement use in the 3 months before conception and the first 6 weeks of pregnancy	NTD
Bower <i>et al.</i> , 2006 ²⁴	Case-control, 1053	Australia	Cases: women who had infants/fetuses with an OFC, CHD, urinary tract defect, limb reduction defect, or other major birth defect; Controls were a random sample selected from the Midwives' Notification System	Self-report of vitamin supplementation periconceptionally, consumption of 174 foods and beverages (6 months before pregnancy), and consumption of any of a list of specific foods fortified with folic acid (6 months before pregnancy)	Birth defects other than NTD
Bukowski <i>et al.</i> , 2009 ²⁶	Prospective cohort, 34 480	US	Women who had singleton births at 20–42 weeks gestation	Self-report (during the first trimester) of preconceptional folate supplementation with or without multivitamins (≥1 year and <1 year)	Pre-eclampsia, SGA, PTD ^a

Table 1. Continued

Reference	Design, <i>n</i>	Location	Population	Intervention/exposure	Outcomes
Burris <i>et al.</i> , 2010 ²⁷	Retrospective cohort, 2464	US	Non-Hispanic white and black mothers of non-malformed infants who participated in the Stone Epidemiology Center Birth Defects Study	Periconceptional multivitamin use ≥ 4 times per week, containing at ≥ 2 water-soluble and two fat-soluble vitamins consumed during the period including the 28 days before and 28 days after the last menstrual period	BW ^a , weight for GA, GA
Catov <i>et al.</i> , 2009 ²⁹	Prospective cohort, 28 601	Denmark	Pregnant women early in gestation	Multivitamin or folate-only supplement use during a 12-week periconceptional period (4 weeks prior 8 weeks after the last menstrual period)	Pre-eclampsia ^a
Catov <i>et al.</i> , 2011 ²⁸	Prospective cohort, 35 897	Denmark	Pregnant women early in gestation	Multivitamin or folate-only supplement use during a 12-week periconceptional period (4 weeks prior 8 weeks after the last menstrual period)	SGA ^a , PTD ^a
De Weerd <i>et al.</i> , 2003 ³⁴	Prospective cohort, 240	the Netherlands	Non-pregnant women who became pregnant during the observation period	Body weight was measured and venous blood samples were assayed for several micronutrients (vitamins A, E, B-1, B-2, B-6, B-12, folate and iron)	Pregnancy loss, BW ^a
Deierlein <i>et al.</i> , 2011 ³⁵	Prospective cohort, 363	US	Mothers of infants who were not preterm and were not diagnosed with illnesses related to infant growth	Self-report of maternal pre-pregnancy BMI (underweight, normal, overweight, obese)	WAZ, LAZ, LWZ
Ehrenthal <i>et al.</i> , 2011 ³⁶	Retrospective cohort, 16 582	US	Women who delivered singleton pregnancies that were not <20 weeks gestation	Self-report of maternal pre-pregnancy BMI (underweight, normal, overweight, obese)	PIH ^a
Han <i>et al.</i> , 2011 ³⁷	Retrospective cohort, 830	South Korea	Pregnant women in their second and third trimesters	Self-report of maternal pre-pregnancy BMI (underweight, normal, overweight, obese)	LBW ^a , PTD ^a
Krapels <i>et al.</i> , 2004 ^{40,41}	Case-control, 409	Netherlands	Cases: mothers of a child with non-syndromic OFC; Controls: women with non-affected child of the same age	Dietary intake according to FFQ 14 months after index birth to estimate preconception diet; Periconceptional vitamin from 4 weeks before and 8 weeks after conception	OFC
Liu <i>et al.</i> , 2011 ⁴³	Retrospective Cohort, 5047	China	Women who had single, nulliparous pregnancies and delivered in one of three hospitals in the Shenyang area	Pre-pregnancy BMI	Pre-eclampsia ^a , PTD, SGA ^a , stillbirth
Murrin <i>et al.</i> , 2007 ⁴⁵	Prospective cohort study, 1048	Ireland	Pregnant women, 14–16 weeks gestation attending their first antenatal visit	Self-report of pre-pregnancy BMI (<25, 25–30, >30)	BW
Oddy <i>et al.</i> , 2009 ⁴⁶	Case-control, 710	Australia	Cases: mothers of infants/fetuses born with a congenital anomaly; Controls: liveborn infants born >20 weeks gestation	Self-report of maternal pre-pregnancy BMI (underweight, normal, overweight, obese)	heart defects, conotruncal defects, NTDs, urinary tract defects, OFCs, limb reduction defects
Ota <i>et al.</i> , 2011 ⁴⁷	Prospective cohort, 3022	Vietnam	Pregnant women with singleton pregnancies	Self-report of pre-gestational BMI (low: <18.5, normal: 18.5–24.9, high: ≥ 25.0)	SGA ^a , large-for-gestational age
Phithakwatchara and Tiapant, 2007 ⁴⁸	Retrospective cohort, 660	Thailand	Pregnant women at risk for gestational diabetes	Pre-pregnancy BMI based on medical records, women with BMI ≥ 27 were compared with women with a BMI of 20–25	Pre-eclampsia ^a , LBW, PTD
Rayco-Solon <i>et al.</i> , 2005 ⁴⁹	Retrospective cohort, 2977	Gambia	Women who had spontaneous vaginal births at home or at the primary care clinic with the assistance of a midwife	Month of birth (determines changes in maternal weight because of fluctuation in rains)	GA ^a
Ronnenberg <i>et al.</i> , 2002 ⁵¹	Case-control, 458	China	Female textile workers who experienced at least one clinically recognised pregnancy during follow up; Cases: ended in fetal death <100 day gestation or livebirth; Controls: ended in a livebirth	Homocysteine, folate, and vitamins B-6 and B-12 concentrations were measured in plasma obtained before conception	Pregnancy loss ^a

Table 1. Continued

Reference	Design, <i>n</i>	Location	Population	Intervention/exposure	Outcomes
Ronnenberg <i>et al.</i> , 2002, 2004 ^{50,54}	Prospective cohort, 405	China	Non-pregnant women textile workers, newly married, aged 21–34	Homocysteine, folate, and vitamins B-6 and B-12 concentrations were measured in plasma obtained before conception	LBW, SGA ^a , BW, IUGR, GA, PTD, infant length, head circumference
Ronnenberg <i>et al.</i> , 2003 ⁵³	Prospective cohort, 575	China	Non-pregnant women textile workers, newly married, aged 21–34	Examined maternal BMI prior to preconception (moderately underweight or severely underweight vs. normal BMI)	BW ^a , LBW ^a , BW ratio, IUGR, GA, PTD, infant head circumference and length, ponderal index
Ronnenberg <i>et al.</i> , 2007 ⁵²	Prospective cohort, 364	China	Non-pregnant women textile workers, newly married, aged 21–34	Homocysteine, folate, and vitamins B-6 and B-12 concentrations were measured in plasma obtained before conception	Pregnancy loss ^a
Shaw <i>et al.</i> , 1995 ⁵⁶	Case-control, 1089	US	Cases: mothers of infants/fetuses with an NTD; Controls: mothers of singleton infants without NTDs	Self-report of vitamin supplement use containing folic acid 3 months before/after conception	NTD ^a
Shaw <i>et al.</i> , 1999 ⁵⁷	Case-control, 829	US	Cases: mothers of infants/fetuses with an NTD; Controls: mothers of singleton infants without NTDs	Nutrient intake according to FFQ, self-report of supplemental vitamin containing folic acid beginning 3 months before conception	NTD ^a , spina bifida, isolated CP ^a
Shaw <i>et al.</i> , 2002 ⁵⁵	Case-control, 1860	US	Cases: mothers of infants/fetuses with a congenital anomaly; Controls: mothers of liveborn infants with no major congenital anomalies	Women's report of use of a vitamin supplement that contained folic acid in the period 1 month before conception through 3 months after conception	Conotruncal defects, isolated CL, CLP, or CP, NTD, limb deficiency defects ^a
Timmermans <i>et al.</i> , 2009 ⁵⁸	Prospective cohort, 6353	the Netherlands	Pregnant women	Self-reported folic acid supplement use (0.4–0.5 mg/day) during first 8 weeks of pregnancy	LBW ^a , SGA ^a , GA, PTD, CHDs
van Beynum <i>et al.</i> , 2010 ⁵⁹	Case-control, 3012	the Netherlands	Cases: mothers of infants with isolated or complex heart defects, without any related syndrome or genetic abnormality; Controls: mothers of infants with a known chromosomal or genetic defect or with other non-folate related congenital malformations	Self-report of folic acid supplementation (≥ 400 µg/day) starting at least 4 weeks prior to conception and continuing up to 8 weeks thereafter	
van Driel <i>et al.</i> , 2008 ⁶⁰	Case-control, 281	the Netherlands	Cases: families with children with a CHD; Controls: families with children without a congenital malformation or chromosomal abnormality	Self-report of periconceptual use of folic acid supplements and completion of a validated FFQ on dietary folic acid and vitamin B-2 intake 17 months after index pregnancy	CHD
Velie <i>et al.</i> , 1999 ⁶¹	Case-control, 859	US	Cases: mothers with NTD-affected infants/fetuses; Controls: randomly selected non-malformed infants	Self-report of preconceptional use of vitamin, mineral, and food supplements and completion of a FFQ	NTD
Vujkovic <i>et al.</i> , 2007 ⁶³	Case-control, 381	the Netherlands	Cases: mothers of a child with CL or CLP; Controls: mothers of non-malformed children	Maternal nutritional intakes assessed 14 months after index birth to estimate preconception intake	CL, CLP ^a
Werler <i>et al.</i> , 1993 ⁶⁴	Case-control, 3078	US and Canada	Cases: mothers of infants/fetuses with NTD; Controls: mothers of infants/fetuses with other major malformation, excluding OFCs	Self-report of multivitamin use containing ≥ 2 vitamins, one which was water soluble taken 28 days before and after conception	NTD ^a
Yazdy <i>et al.</i> , 2007 ⁶⁵	Pre-post, 420 832	US	Women giving birth in 54 states in the US (and DC)	Impact of folic acid fortification (pre vs. post-fortification)	OFC ^a
Yeh and Shelton, 2007 ⁶⁶	Retrospective cohort, 1342	US	Women who delivered live-born twins in Western upstate New York	Maternal pre-pregnancy BMI (underweight, normal, overweight, obese) based on information from birth certificates	mean twin BW, GA

^aReported significant associations.

BMI, body mass index; BW, Birthweight; CHD, congenital heart defects; CL, cleft lip; CLP, cleft lip with cleft palate; CP, cleft palate; FFQ, food frequency questionnaire; GA, gestational age; IUGR, intrauterine growth restriction; LAZ, length-for-age z-score; LBW, low Birthweight; LWZ, length-for-weight z-score; PTD, neural tube defects; OFC, oro-facial cleft; PIH, pregnancy induced hypertension; PTD, preterm delivery; RCT, randomised controlled trial; SGA, small-for-gestational age; WAZ, weight-for-age z-score.

risk ratio (RR): 5.7 [95% CI 4.0–8.1] and 3.0 [95% CI 2.2–4.1] of pre-eclampsia. Ehrental *et al.*³⁶ also found that prepregnancy BMI (based on self-report) was positively associated with the risk of pregnancy-induced hypertension.

In a secondary analysis of a large prospective cohort study, Bukowski *et al.*²⁶ found no significant association between duration of preconceptional folate supplementation and the risk of pre-eclampsia (OR: 1.05 [95% CI 0.85, 1.31]) or placental abruption (OR: 0.80 [95% CI 0.55, 1.14]). In contrast, regular consumption of multivitamin supplements during the periconceptional period was associated with a 22% reduced risk of pre-eclampsia [hazards ratio (HR): 0.78 [95% CI 0.60, 0.99]] in a study using data from the Danish National Birth Cohort (1997–2003).²⁹

Pregnancy loss

De Weerd *et al.*³⁴ evaluated the relation between maternal periconceptional biochemical and haematological parameters and early pregnancy loss in a prospective study of 240 women in the Netherlands. Women were recruited before pregnancy and body weight measurements and blood samples were taken preconceptionally and at 6 and 10 weeks amenorrhea. Prepregnancy weight was positively associated with the risk of early pregnancy loss ($P < 0.05$) but relationships with concentrations of several biomarkers of vitamin status were non-significant in this well-nourished population. In contrast, findings from an observational study of Chinese women textile workers suggest that preconception micronutrient status may be negatively associated with pregnancy loss.^{51,52} Ronnenberg *et al.*⁵¹ found that suboptimal preconception folate and vitamin B-6 status, especially when they occurred together, was associated with an increased risk of clinical spontaneous abortion (P for trend = 0.06 and 0.07, respectively) in a case-control analysis in which cases ($n = 49$) were women with a clinically recognised pregnancy who experienced a fetal death before 100 days' gestation and controls ($n = 409$) were women who maintained a pregnancy that ended in a livebirth. Ronnenberg *et al.*⁵² also found that compared with women in the lowest quartile of vitamin B-6 levels, those in the third and fourth quartiles were more likely to conceive (adjusted HR: 2.2 [95% CI 1.3, 3.4], HR: 1.6 [95% CI 1.1, 2.3], respectively), and the risk of early pregnancy loss in concep-

tive cycles was lower in the fourth quartile (OR: 0.5 [95% CI 0.3, 1.0]). This analysis was done in the subsample of 364 women who conceived at least once during the period 1996–1998. Liu *et al.*⁴³ did not find any significant differences in the risk of stillbirth by categories of prepregnancy BMI.

Five intervention trials^{33,38,39,42,44} evaluated the effect of periconceptional folic acid on miscarriages and/or stillbirths and found no significant differences. These trials were conducted primarily in developed countries among women at risk of delivering a child with NTD and have been described below. One trial that was conducted in Algeria, evaluated the effect of providing iodised oil in women either before or during the first trimester of pregnancy and reported a non-significant reduction in the incidence of stillbirths when compared with women who received no intervention.³⁰

Birth defects

We found eight intervention trials^{20,31–33,38,39,42,44,62} and 14 observational^{23–25,40,41,46,55–57,59–61,63–65} studies that evaluated the relationship between maternal preconceptional and periconceptional nutrition and risk of birth defects, especially NTD. Czeizel *et al.*^{32,33} compared the risk of NTD births among women receiving vitamin supplement (containing 0.8 mg folic acid) and those receiving trace-element supplements daily from at least 1 month before conception and until the date of the second missed menstrual period or later in a randomised controlled trial (RCT) among 7540 Hungarian women (<35 years) and showed significant reductions ($P < 0.05$) in congenital malformations (13.3/1000 and 22.9/1000 in the vitamin and trace-element group, respectively) and the first time occurrence of NTD (6 vs. 0 NTD cases in the trace-element and vitamin supplement group). The MRC Vitamin Study Group⁴⁴ also evaluated the effects of supplementation with folic acid or a mixture of seven other vitamins (A, D, B-1, B-2, B-6, C and nicotinamide) around the time of conception in a large multicenter double blind RCT of 1817 women in the UK and other countries and reported a 72% reduction in the incidence of NTDs in the folic acid group (RR: 0.28 [95% CI 0.12, 0.71]) but no significant protective effect for the other vitamins group (RR: 0.80 [95% CI 0.32, 1.72]) when compared with a placebo. Intervention trials in China, India and Ireland have shown similar results.^{20,31,38,39}

Observational studies have also shown the protective effect of folic acid in reducing birth defects. Shaw *et al.*⁵⁶ found that women who consumed a folic acid-containing supplement in the 3 months before conception had a lower risk of having an NTD-affected pregnancy (OR: 0.65 [95% CI 0.45, 0.94]) when compared with non-users. Werler *et al.*⁶⁴ also found that daily periconceptual intake of 0.4 mg of folic acid was associated with a 60% reduction in the risk of occurrent NTDs in a case-control study where cases were mothers of infants/fetuses with NTD while controls consisted of mothers of infants/fetuses with other major malformation, excluding oro-facial clefts (OFC). The exposure was defined as consumption of multivitamin supplements containing ≥ 2 vitamins including folic acid 28 days before and after conception. In contrast, Bower and Stanley²⁵ did not find an association between periconceptual vitamin supplementation and NTDs in a case-control study in Australia. Recall bias and small sample size may have been limitations. We also found a case-control study⁶¹ in which higher preconceptional zinc intake was associated with a reduced risk for NTD (quintile 5 vs. quintile 1, OR: 0.65 [95% CI 0.43, 0.99]) cases were 430 NTD-affected fetuses/infants, and controls were 429 randomly selected non-malformed infants.

A few studies have examined the relationship between preconceptional nutrition and other birth defects such as cleft palate and congenital heart defects (CHD). Vujkovic *et al.*⁶³ assessed maternal preconceptional nutritional intakes in a case-control study of 203 mothers of a child with a cleft lip or cleft palate and 178 mothers with non-malformed offspring and found that the Western dietary pattern, for example, high in meat, pizza, legumes and potatoes, and low in fruits, was associated with a high risk of a cleft lip or cleft palate (OR: 1.9 [95% CI 1.2, 3.1]). Bower *et al.*²⁴ found no association between folic acid supplement use during the periconceptual period and risk of birth defects other than NTD. Cases were women whose infants had OFC ($n = 62$), CHD ($n = 151$), urinary tract defects ($n = 117$), limb reduction defects ($n = 26$) or other major birth defects ($n = 119$). There were 578 control women. Van Driel *et al.*⁶⁰ in a case-control study of Dutch women also found no significant differences in periconceptual use of folic acid supplements and dietary intakes of total energy, folate, and vitamin B-2 between case (infants with CHD) and control-mothers, but reported a significant interaction

between genetics and folic acid supplement use during the periconceptual period ($P = 0.008$); the OR [95% CI] of the mothers carrying the MTHFR AC and CC genotypes in the supplemented vs. the non-supplemented group was 1.8 [95% CI 1.01–3.1] vs. 0.6 [95% CI 0.3–1.1], respectively. A case-control study in the Netherlands^{40,41} also showed a trend towards risk reduction for OFC with increasing dietary intake of thiamine ($P = 0.04$) and pyridoxine ($P = 0.03$) among women who consumed folic acid supplements periconceptionally. Finally, Yazdy *et al.*⁶⁵ reported a significant decline in OFC prevalence following folic acid fortification (prevalence risk: 0.94 [95% CI 0.92, 0.96]) based on retrospective cohort analysis of US birth certificate data from 45 states and the District of Columbia in which births were compared during the pre-fortification period (January 1990–December 1996) and post-fortification period (October 1998–December 2002).

Birth size

We identified two intervention trials^{30,33} and 14 observational studies^{26,27,34,35,37,43,45,47,48,50,53,54,58,66} that examined the relationship between prepregnancy and/or periconceptual nutrition and birth size. Chaouki and Benmiloud³⁰ evaluated the benefits of providing oral iodised oil to women just before conception or during the first trimester in a study conducted in a region of endemic goiter in Algeria. The offspring of treated women ($n = 1536$) had significantly higher birth-weight (+6.25%) when compared with non-treated women. Czeizel *et al.*³³ found no significant differences in the risk of LBW when they compared women who received folic acid containing supplements before 12 weeks gestation with those who received a supplement containing only trace elements (copper, manganese and zinc) and vitamin C. Mean birth-weight was much higher in both groups compared with the general population.

Several observational studies have examined the association with maternal nutritional status based on anthropometric measurements such as weight and height and/or vitamin supplement use during the periconceptual period and birth size. Liu *et al.*⁴³ reported an increased risk of delivering a SGA infant (adjusted RR: 1.7 [95% CI 1.1, 2.6]) among underweight women ($\text{BMI} < 18.5 \text{ kg/m}^2$) in a retrospective study of Chinese women. Ronnenberg *et al.*⁵³ reported similar findings in a prospective cohort study that

examined the relationship between pre-pregnancy BMI and birth outcomes among 20- to 34-year-old Chinese women ($n = 575$); infants born to mothers who were underweight before pregnancy ($\text{BMI} \leq 18.5 \text{ kg/m}^2$) were at increased risk for fetal growth deficits. Being underweight was also associated with smaller infant head circumference and lower ponderal index. Prepregnancy weight ($P < 0.01$; partial $r^2 = 0.24$) was positively associated with infant birthweight in a prospective study of 240 women in whom measurements were obtained preconceptually in the Netherlands.³⁴ Most recently, a large prospective study from Vietnam⁴⁷ also reported a significantly higher risk of delivering a SGA infant (adjusted OR: 1.95 [95% CI 1.52–2.50], $P < 0.01$) among women who were underweight before conception ($\text{BMI} < 18.5 \text{ kg/m}^2$) compared with those with BMI between 18.5 and 23.0 kg/m^2 . There were no significant differences for the group with higher BMI ($>23 \text{ kg/m}^2$). Similarly, no significant differences were reported in the risk of LBW (OR: 0.57 [95% CI 0.29, 1.14]) by maternal overweight ($\text{BMI} \geq 27 \text{ kg/m}^2$) in a retrospective study of Thai women.⁴⁸

Ronnenberg *et al.*⁵⁴ also assessed the association between preconception anaemia and iron status and infant growth and pregnancy outcomes and found that preconception anaemia, particularly iron-deficiency anaemia, was associated with reduced infant growth (lower birthweight). The risks of LBW and fetal growth restriction (defined as $<85\%$ of a birthweight ratio calculated as the observed birthweight*100/mean birthweight of infants with the same gestational age within the cohort) were significantly greater among women with moderate anaemia compared with non-anaemic controls (OR: 6.5 [95% CI 1.6, 26.7], $P = 0.009$ and OR: 4.6 [95% CI 1.5, 13.5], $P = 0.006$, respectively). A few studies, primarily in developed countries have also examined the relationship between periconceptual multivitamin use and birth size. In a retrospective cohort study of non-Hispanic white ($n = 2331$) and non-Hispanic black ($n = 133$) mother–infant pairs, Burris *et al.*²⁷ assessed the association of maternal periconceptual multivitamin use and infant birthweight disparities between infants delivered by whites and those delivered by their African American women counterparts. Multivitamin use was associated with a 536 g increased birthweight ($P = 0.001$) among African Americans; no association between multivitamin use and birthweight or gestational age was found among white subjects.

This study used a more restrictive definition of periconceptual as the period between 28 days prior to the date of the last menstrual period to 28 days after last menstrual period. In a population-based prospective cohort study, Timmermans *et al.*⁵⁸ evaluated the impact of self-reported folic acid supplement (0.4–0.5 mg/day) and found that periconceptual folic acid supplementation (<8 weeks of gestation) was associated with higher placental (13 g [95% CI 1.1, 25.5]) and birthweight (68 g [95% CI 37.2, 99.0]); reduced risks for LBW (OR: 0.43 [95% CI 0.28, 0.69]) and SGA (OR: 0.40 [95% CI 0.22, 0.72]) were observed for women who started supplementation preconceptually, compared with those who did not use folic acid. A significant interaction by parity was also observed, with larger differences in birthweight by supplement use among multiparous compared with nulliparous women. The adjusted risk for a SGA birth was also significantly reduced among regular users of multivitamins during the periconceptual period (<12 weeks gestation) regardless of their prepregnancy BMI (HR: 0.83 [95% CI 0.73, 0.95]) compared with non-users in the Danish National Birth Cohort Study ($n = 35\,897$).²⁸

Preterm delivery/gestational age

We found two intervention trials^{30,33} and 11 observational studies^{26–28,37,43,48–50,53,54,58,66} that examined the relationship between preconceptual or periconceptual nutrition and gestational age and/or risk of PTD. There were no significant differences in the incidence of PTD in the large intervention trial in Hungary that examined the benefits of providing folic acid containing supplements before 12 weeks of gestation, but both groups received other micronutrients like zinc. Chaouki and Benmiloud³⁰ found no differences in PTD among women who received iodised oil during the first trimester of pregnancy.

Rayco-Solon *et al.*⁴⁹ evaluated the effect of pre-conceptual undernutrition among rural Gambian women who experience annual fluctuations in energy balance and found significantly shorter gestational ages for pregnancies conceived in September to November (when lower weights are recorded) than those from better-fed months (38.6 vs. 39.0 weeks; log-rank $\chi^2 = 17.4$, $P < 0.0001$). Data were obtained prospectively in this study. Liu *et al.*⁴³ obtained measures of body size (weight and height) before 12 weeks gestation from health records and found that the adjusted

odds of early PTD (<34 weeks) was significantly elevated (RR: 3.4 [95% CI 1.2, 9.4]) among obese women (BMI > 28 kg/m²) when compared with normal weight women (18.5 < BMI < 24); there were no differences in the risk of PTD (<37 weeks) by maternal prepregnancy size in contrast to findings from earlier observational studies.^{37,48,53,58,86–88} Han *et al.*³⁷ found that high pre-pregnancy maternal BMI (>25 kg/m²) increased the risk of PTD (OR: 2.85 [95% CI 1.20, 6.74]) in a study of Korean women.

Using a case-control design, Ronnenberg *et al.*⁵⁰ found that elevated homocysteine (≥ 12.4 μ mol/L) during the preconceptional period was associated with a higher risk of PTD (OR: 3.6 [95% CI 1.3, 10.0], $P < 0.05$). The risk of PTD was also 60% lower among women with serum vitamin B-12 ≥ 258 pmol/L than among vitamin B-12-deficient women (OR: 0.4 [95% CI 0.2, 0.9], $P < 0.05$). Similar reductions were seen with among vitamin B-6-deficient women (OR: 0.5 [95% CI 0.2, 1.2]), but they were not statistically significant. Folate status was not associated with PTD. In contrast, Bukowski *et al.*²⁶ reported that preconceptional folate supplementation that was prospectively recorded in the first trimester of pregnancy was associated with significant reductions in the incidence of early spontaneous PTD (HR: 0.22 [95% CI 0.08, 0.61], $P = 0.004$). Regular periconceptional multivitamin use was associated with reduced risk of PTD in non-overweight women (HR: 0.84 [95% CI 0.73, 0.95]) who participated in the Danish Birth Cohort Study.²⁸ We did not find any reports of controlled trials in developing country populations where there is greater risk of inadequate nutrient intakes.

Comments

The majority of studies identified in our review were observational with few well-designed intervention trials. The overall quality of evidence was low for most outcomes with the exception of the benefits of maternal preconception folic acid for reducing the risk of NTD, which was high (Table 2) and based on several well-designed intervention trials. The most recent meta-analysis by De-Regil *et al.*⁸⁹ that included five intervention trials^{32,33,38,39,42,44} confirmed the findings of an earlier review¹⁸ that estimated a 72% reduction in the risk of NTD (RR: 0.28 [95% CI 0.13, 0.58]). These findings have led to recommendations promoting the use of supplements and/or fortified foods containing folic acid by women of reproductive age

(WRA) in many countries. The universal fortification of staple foods such as flour and ready-to-eat cereals in countries such as the US, Canada, Chile and Costa Rica has been linked to significant reductions in the incidence of NTD in these countries.^{90,91} Some of the studies we reviewed also suggest that increased intakes of folic acid and other nutrients are associated with reduced risk of other congenital birth defects.^{40,41,63,65} Unpublished findings based on follow-up of offspring in the SINO-US NTD prevention project have shown that daily consumption of 400 μ g of folic acid during the periconceptional period was associated with reduced infant mortality among infants without major birth defects (RR = 0.78 [95% CI 0.72, 0.85]) and improved linear growth.^{92,93} There were no differences in behaviour and cognitive development during early childhood in the same study population^{94,95} in contrast to a recent study in which maternal consumption of supplements containing folic acid during the periconceptional period (4 weeks before pregnancy to <8 weeks gestation) was associated with a significant reduction (20%) in the risk of moderate language delay among the offspring (singleton non-intrauterine growth restriction) at 3 years of age in a Norwegian birth cohort.⁹⁶ Finally, no benefit was reported in a recent meta-analysis of four trials that examined the effects of were periconceptual folic acid supplementation on stillbirths (RR: 0.96 [95% CI 0.51, 1.83]); all trials were conducted among women with a history of NTD in a previous pregnancy.⁸⁹

Overall, we found few studies from developing country settings where maternal malnutrition is common. Short interpregnancy interval, which has the potential to result in maternal depletion of nutrients including folate, iodine and iron, has been associated with increased risk of adverse outcomes such as fetal growth restriction and developmental abnormalities.^{13,14} For example, severe iodine deficiency during pregnancy has been associated with adverse pregnancy outcomes including cretinism⁹⁷ and although we found a few intervention studies^{30,69,74} that suggest that providing iodine during early pregnancy to women living in iodine deficient areas improve birth size, we had to exclude some of them because the exact nature and timing of the intervention was unclear. Universal salt iodisation has been a successful strategy towards the elimination of iodine deficiency disorders but the importance of iodine in settings where mild-moderate iodine deficiency exists has not been studied adequately and appropriate intervention

Table 2. Quality assessment and summary of findings

No. of studies	Study design	Quality assessment			Qualitative summary of findings
		Limitations	Consistency	Generalisable to resource-poor populations	Generalisable to intervention of interest
Pre-eclampsia: overall quality of evidence = very low 4 Three prospective cohort, one retrospective cohort		Unclear if measures of exposure were restricted to preconception and/or the first trimester of pregnancy only; no intervention trials	No	U5, Denmark and overweight Thai women at risk for gestational diabetes; China	One of two large studies showed reductions in pre-eclampsia in multivitamin users; one study from China showed positive associations with overweight
Pregnancy loss: overall quality of evidence = low 10 Six intervention trials		Blinding unclear for some trials; studies among women with history of NTD	No	Algeria (high risk of endemic goiter) Ireland, Hungary, UK, India and China the Netherlands, China	No significant differences for in stillbirth Five of six studies provided multivitamins containing folic acid Biomarkers of nutritional status (water soluble vitamins and iron) in samples collected just before and/or in early pregnancy Prepregnancy BMI
Birth defects: overall quality of evidence = high 22 Eight intervention trials	Four observational studies • Two prospective cohort • One retrospective cohort • One case-control	Recall and selection bias	Yes	Ireland, Hungary, UK, India, China	One prospective study in China found associations between vitamin status and pregnancy loss whereas the one from Netherlands found no associations. One study showed no differences in stillbirth by maternal BMI.
Birthweight (g): overall quality of evidence = low 7 Two intervention trials	14 Case-control Recall bias and small sample size for some studies	Blinding unclear for study in Algeria and allocation was broken early in Hungarian RCT Exposures varied from prepregnancy body size to micronutrient status	No	Many studies in North America, Europe, Australia, and China Algeria (has endemic goiter), Hungary (low risk of deficiencies) the Netherlands, US, China, Ireland	72% reduction in NTD among women who consumed folic acid containing supplements during the periconceptional period Mixed findings with suggestion of benefit of NTD and other birth defects as well
LBW (<2500 g): overall quality of evidence = low 6 One intervention trial	Five observational studies • Three prospective cohort • Two retrospective cohort	Allocation was broken early in Hungarian RCT Recall and selection bias especially in retrospective studies. Exposures varied from prepregnancy body size to micronutrient status.	No	Hungary (low risk of deficiencies) Korea, Thailand, Netherlands, China	Significantly higher birthweight for iodine but no differences for multivitamin group but control group also received selected nutrients (Cu, Zn, Iron and Mg) Underweight is positively associated with birthweight
SGA: overall quality of evidence = low 8 Three retrospective cohort	Five prospective cohort Three retrospective cohort	Recall and selection bias especially in retrospective studies. Varying definitions for exposures and outcomes.	No	China, Vietnam, US, Netherlands, Denmark	No significant differences All prospective studies suggest benefit of periconceptional micronutrient status or supplement use Periconceptional use of multivitamin supplements is associated with reduced risk of SGA in developed countries. Low BMI (<18.5 kg/m ²) before or during early pregnancy is associated with increased risk of SGA in prospective studies.
Gestational age (weeks): overall quality of evidence = low 7 One intervention trial	Six observational studies • Three prospective cohort • Three retrospective cohort	Allocation was broken early in Hungarian RCT Recall and selection bias especially in retrospective studies. Few well-designed studies by specific exposure (BMI, supplement use and micronutrient).	No	Hungary (low risk of deficiencies) US, China and Gambia	No differences Mixed findings
Preterm delivery: overall quality of evidence = low 8 Two intervention trials	Six observational studies • Three prospective cohort • Three retrospective cohort	Blinding unclear for study in Algeria and allocation was broken early in Hungarian RCT Recall and selection bias especially in retrospective studies	No	Algeria (has endemic goiter), Hungary (low risk of deficiencies) US, the Netherlands China and Korea, Thailand	No differences in both trials Two of three prospective studies show benefit of periconceptional micronutrient supplement. Two studies indicate increased risk among overweight women.

BMI, body mass index; LBW, low Birthweight; NTD, neural tube defects; RCT, randomised controlled trial; SGA, small-for-gestational age.

in WRA may have the potential to improve MNCH outcomes. Several intervention trials have also shown that the provision of weekly iron-folic acid supplements to WRA in developing country populations can improve iron status and reduce the risk of anaemia^{73,75,76,78–80,82} but few have evaluated the benefits of these interventions for pregnancy outcomes. This is a major gap given the high prevalence of iron deficiency and anaemia in WRA (before and during pregnancy) in many developing countries and our understanding of the mechanisms that suggest that iron status during the periconceptual period may be as important as iron status during the latter half of pregnancy for improving MNCH outcomes.³ We found only one observational study that showed that anaemia during the preconception period in WRA is associated with increased risk of unfavourable pregnancy outcomes and reduced infant growth.⁵⁴ The major cause of anaemia is iron deficiency, however, deficiencies of other micronutrients such as folate, vitamins B-6 and B-12 can also cause anaemia, indicating that the inclusion of these nutrient may be beneficial. Periconceptual anaemia may influence the synthesis of hormones and thus adversely affect infant growth.^{3,98} The current review, including mostly observational studies in developed countries, suggests that preconceptional and periconceptual intake of vitamin and mineral supplements or dietary intake of such nutrients may reduce the risk of adverse outcomes such as PTD and LBW.

We found few well-designed studies that carefully examined the relationship between maternal size and body composition during the periconceptual period with adverse pregnancy outcomes such as stillbirth, PTD and LBW. Recent reviews and meta-analyses have concluded that balanced protein-energy supplementation during pregnancy was associated with reduced stillbirth rates and LBW, but none of the studies examined the effect of these interventions before and during the first trimester of pregnancy.⁹⁹ Several observational studies especially in developed countries have examined the relationship between maternal BMI and adverse pregnancy outcomes, but most of them used data that were obtained after delivery or based on medical records obtained during pregnancy making it difficult to ascertain the quality of the data and exact timing of measurement.^{12,86,88,100} Nevertheless, these studies do suggest that overweight and/or obesity is associated with increased risk of pregnancy complications such as gestational

diabetes and hypertensive disorders which in turn influence subsequent maternal and child health and well-being.⁸⁷ The Institute of Medicine, which recently revised the US guidelines for gestational weight gain (GWG) based on the concerns about the obesity epidemic, also concluded that the risk of SGA was greater among women who had low GWG and low prepregnancy BMI and that there was strong evidence of a U shaped relationship between low GWG and PTD in normal and underweight women.¹¹ These findings have important policy implications for industrialised countries such as the US as well as countries like Mexico that are facing the dual burden of malnutrition, that is, undernutrition that manifests as stunting during early childhood and contributes to short maternal stature combined with overweight as a result of the nutrition transition. Careful examination of the relationship between prepregnancy body size and composition and MNCH outcomes in developing country settings using well-designed prospective cohort studies is needed to develop appropriate interventions.

Overall, the paucity of intervention trials, especially in developing country settings is striking. The majority of the studies in our review were observational in design, which make inferences of causality difficult especially when the exposure was based on maternal recall. Only a third of the studies measured the exposure, that is, nutritional status and/or intakes using a prospective cohort study design. Key limitations include the inability to determine if the outcomes are specifically the result of preconceptional supplementation because women who consumed nutrient supplements before and during early pregnancy continued to take them through delivery, recall bias (in the case of retrospective studies) and differences in the timing of exposure. There is no clear definition of the 'periconceptual' period; we used <12 weeks, that is, first trimester which has been used by others and typically in many settings especially developing countries, most women do not identify and/or seek antenatal care before 12 weeks. Some studies clearly state that they measured nutrient intakes and/or status before pregnancy (maternal recall or prospectively) but the periconception period ranged from 1 month prior to the last menstrual period to 4 to 12 weeks of gestation. We also excluded several studies because they either used a slightly different definition, namely <16 or 20 weeks gestation, and/or the timing of the intervention was not clear.

In summary, there is evidence supporting the importance of nutritional status before and during early pregnancy to reduce the risk of adverse pregnancy outcomes especially birth defects and to a lesser extent, PTD and LBW. Little is known about outcomes such as stillbirths and maternal and infant mortality. The limited available evidence suggests improving prepregnancy maternal nutritional status will improve MNCH outcomes, although there are emerging concerns of overweight and obesity. There is a need for RCTs that evaluate the benefits of preconceptional nutritional interventions to confirm the findings from observational studies. These studies will need large sample sizes and should evaluate interventions such as providing supplements that contain nutrients such as iron, zinc, iodine and/or a combination of several micronutrients in addition to providing folic acid, targeted use of fortified foods and or behaviour modification to improve intakes. The dissemination of messages about the importance of a healthy diet and lifestyle before and during pregnancy along with messages about family planning that address timing and spacing of pregnancies have the potential to optimise MNCH outcomes in many settings. Evaluation of innovative approaches such as counselling newly wed mothers¹⁰¹ is also lacking and will help guide programme implementation.

Conflicts of interest

The authors declare no conflicts of interests.

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